

Lecture 8

Arousal & Sleep

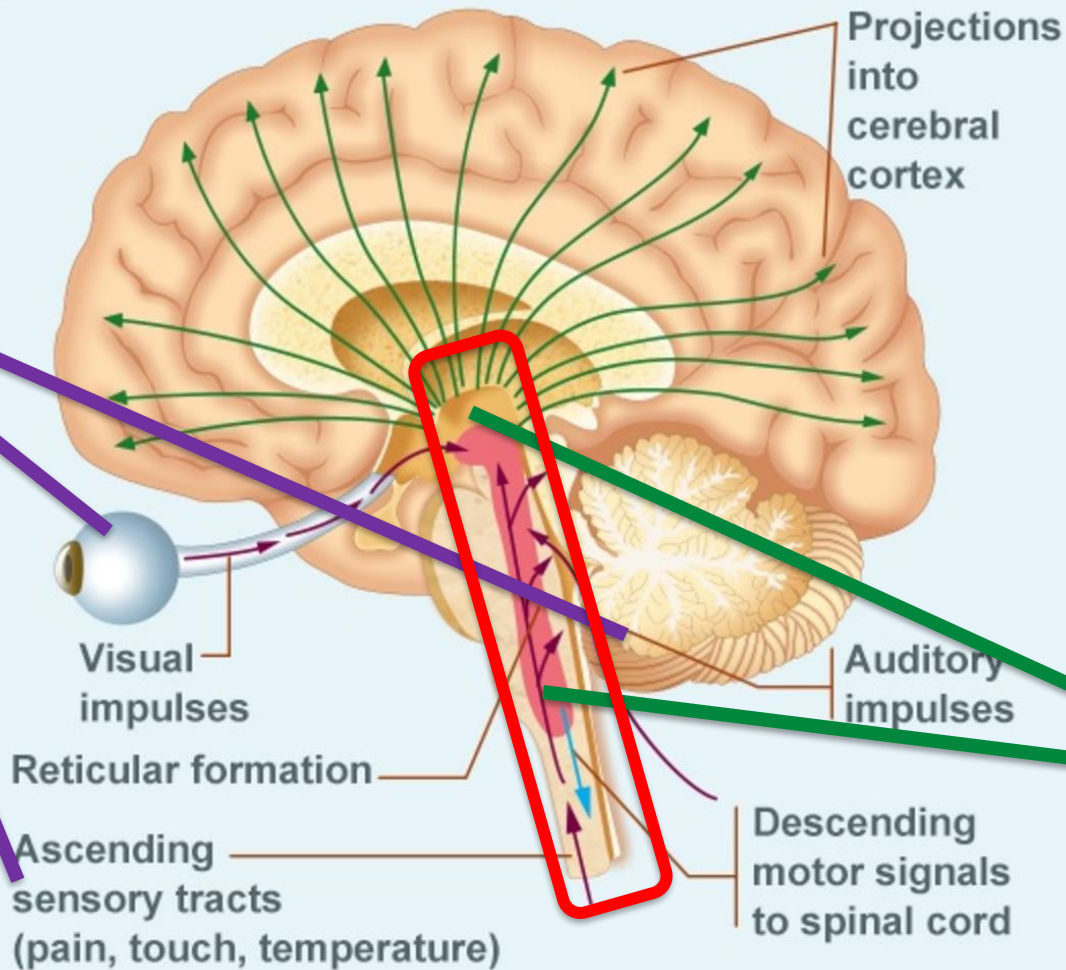


Arousal in the Brain

Reticular Activating System

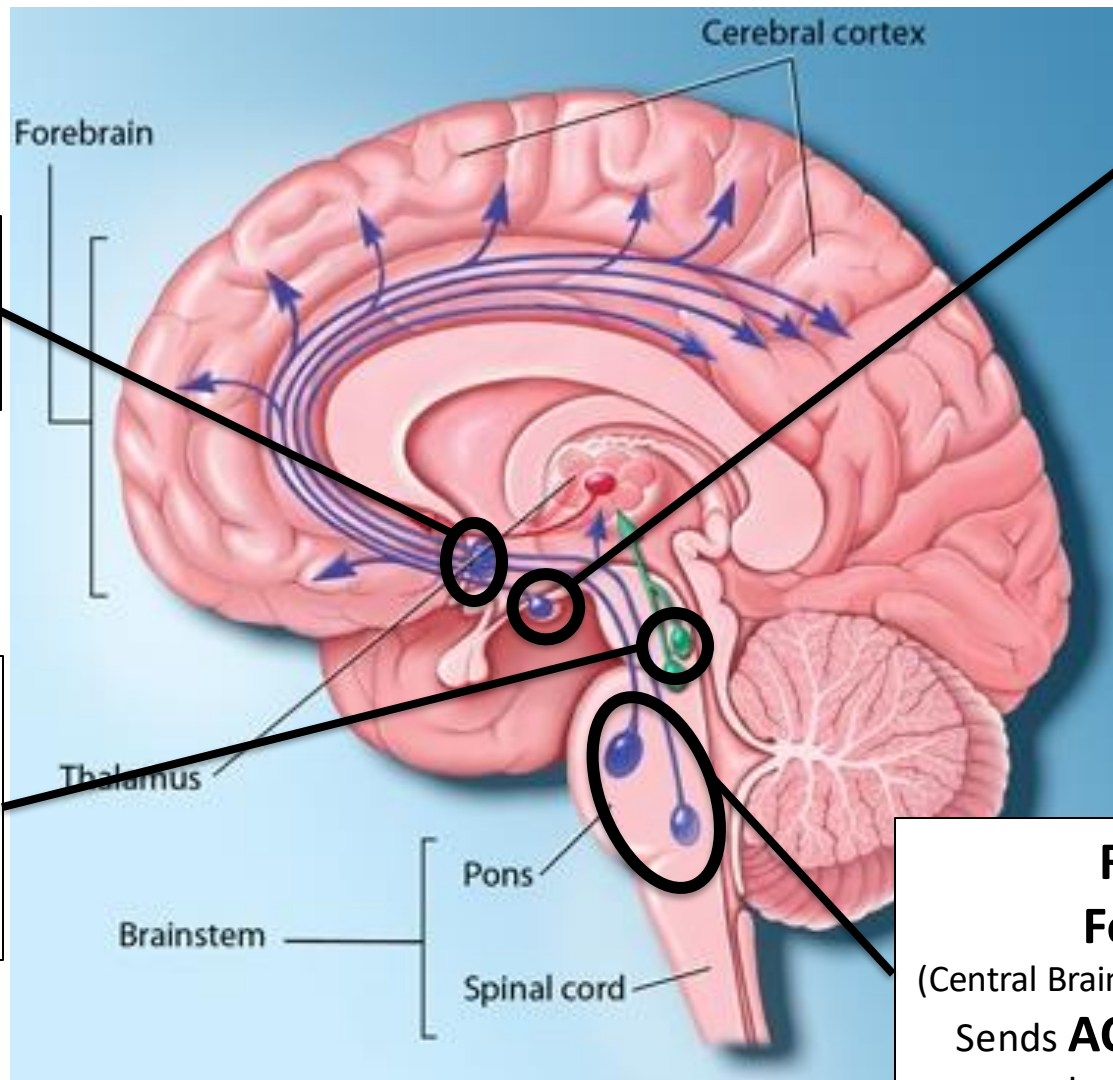
Stimulated by
sensory input

Mnemonic:
R.A.S.
"Rise And Shine"



Also initiated,
maintained
endogenously

Arousal in the Brain



Basal Forebrain

Delivers **ACh** throughout cortex

Lateral Hypothalamus releases **Orexin**, helps maintain arousal in these other systems, based on body conditions

Locus Coeruleus

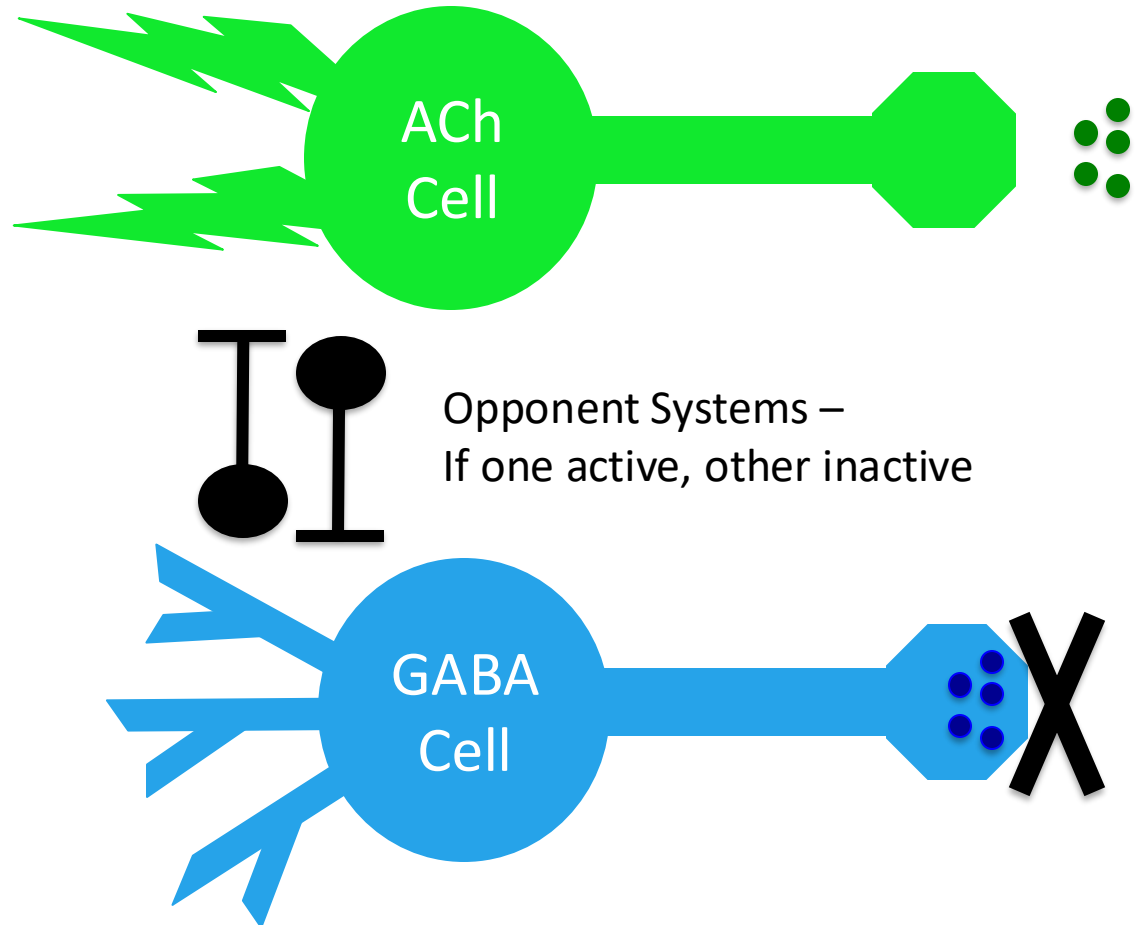
("Dark Blue Place")

Releases **NE** for Vigilance, Alerts memory

Reticular Formation
(Central Brainstem, Lateral to Raphe)
Sends **ACh & Glutamate** throughout brain

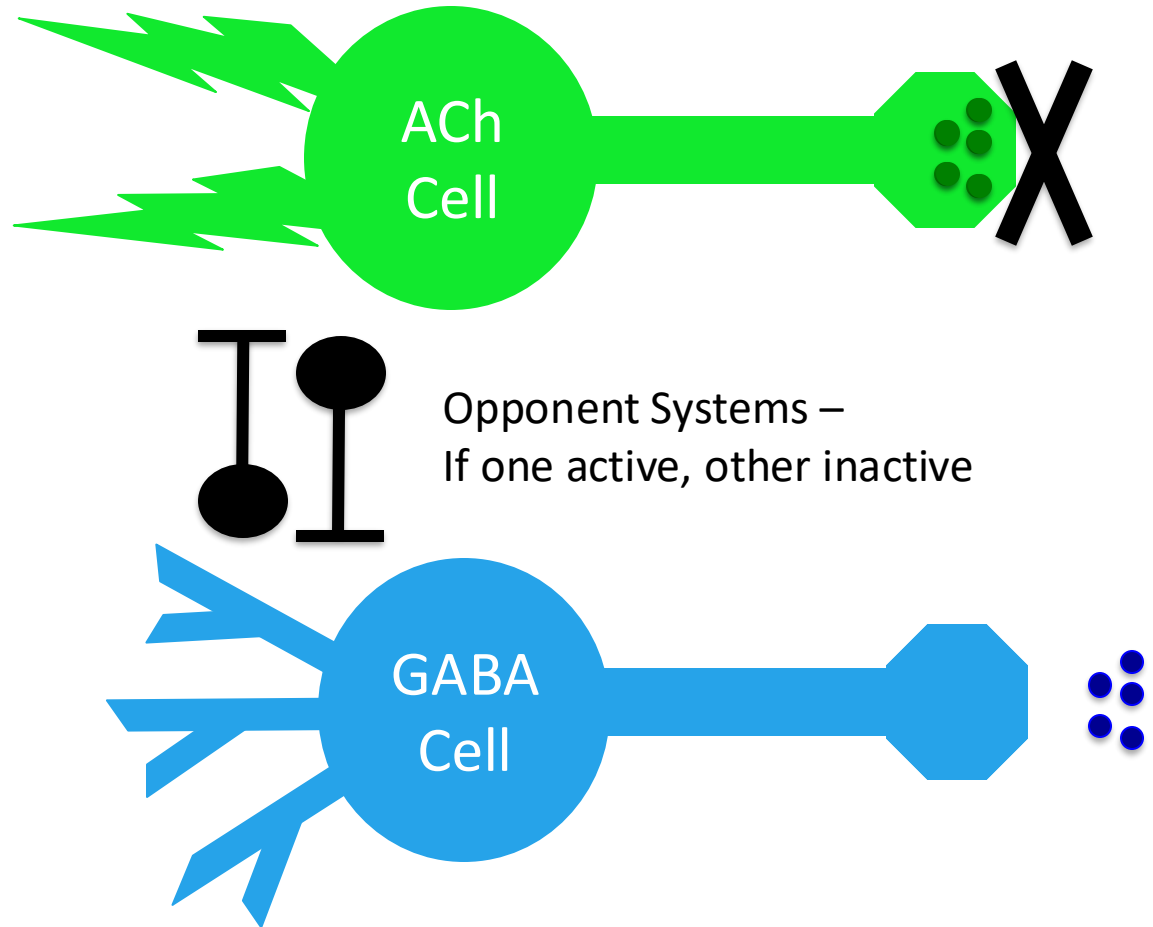
How does Caffeine help you wake up?

Basal Forebrain



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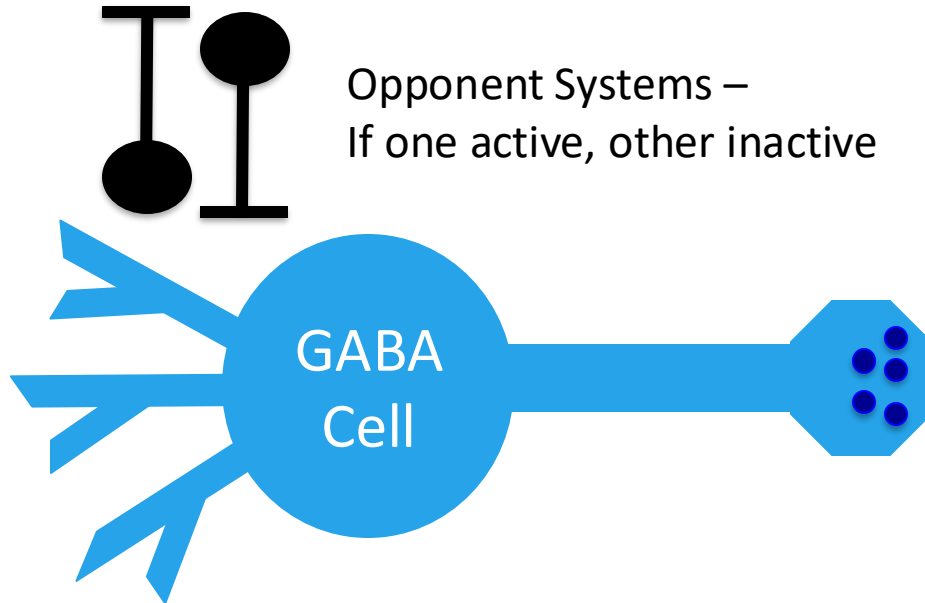
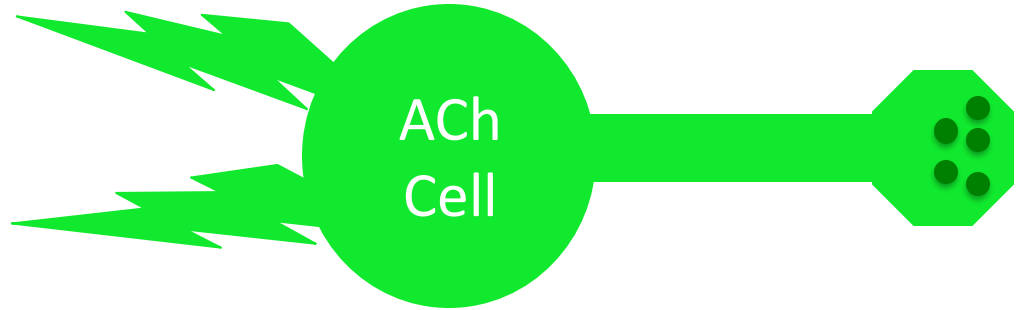
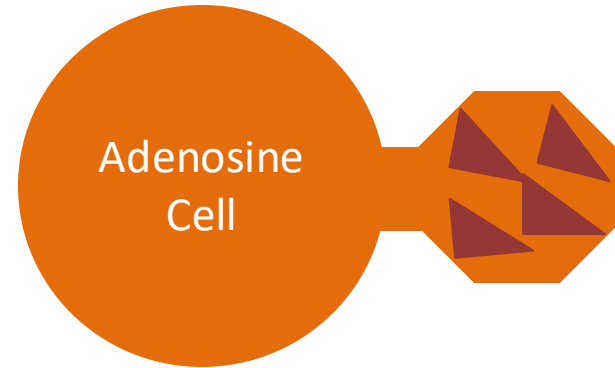
Adenosine
Cell

Adenosine 
(a metabolic byproduct)
builds up in cells
during the day

ACh
Cell

Opponent Systems –
If one active, other inactive

GABA
Cell



How does Caffeine help you wake up?

Basal Forebrain

Adenosine
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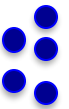
ACh
Cell

GABA
Cell

Adenosine 
(a metabolic byproduct)
builds up in cells
during the day

Its release
Inhibits ACh cells
in Basal Forebrain,
allowing **GABA** system
to dominate
>> **Sleep**

Opponent Systems –
If one active, other inactive



How does Caffeine help you wake up?

Basal Forebrain

NOTE: ACh cell is also simultaneously being excited by other arousal inputs

Adenosine Cell

ACh Cell

Caffeine

(crosses blood-brain barrier)

BLOCKS Adenosine receptor sites

But does NOT inhibit ACh cells

So ACh system active, which suppresses GABA system
>> **Awake!**

Opponent Systems –
If one active, other inactive

GABA Cell

EEG - Electro-Encephalogram



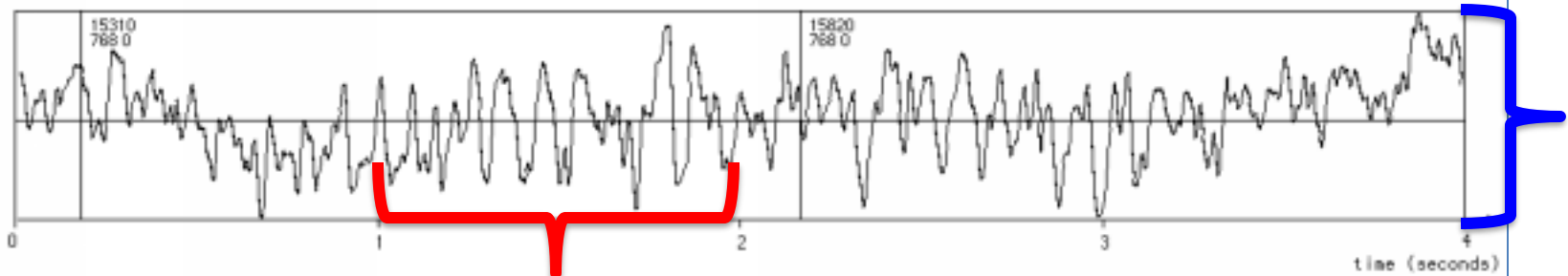
Gross average change in
electrical potentials
in area under electrode

Amplitude = Voltage

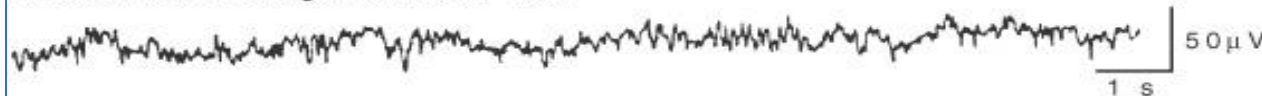
Frequency = Cycles per second (Hz)

Sample EEG Recording

Time series



Awake: low voltage-random, fast



Awake

Beta (18-24 Hz)

Awake: 8 to 12 cps- alpha waves



Alert & Relaxed

Alpha (8 -12 Hz)

Stage 1: 3 to 7 cps- theta waves



Stage 1

Theta (4-7 Hz)

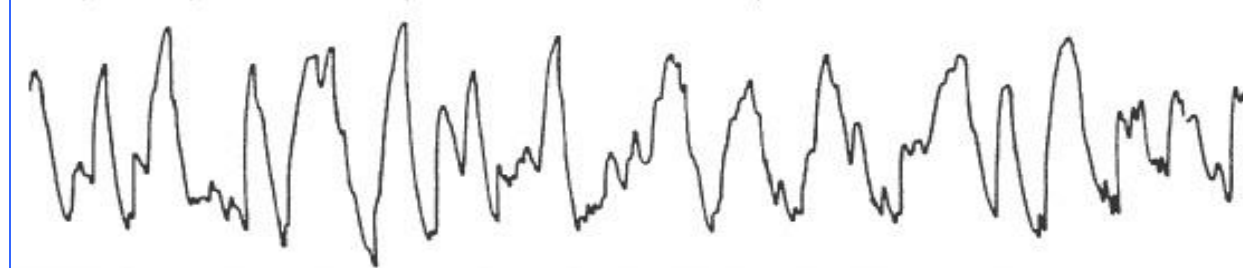
Stage 2: 12 to 14 cps- sleep spindles and K complexes



Stage 2:

Theta w/Spindles
and K-Complexes

Deep sleep: 1/2 to 2 cps- delta waves >75 μV



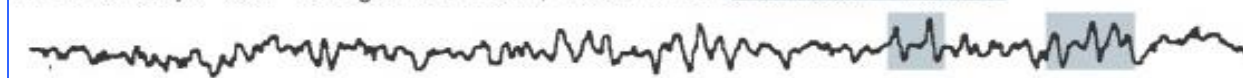
Stage 3 & 4

“Slow Wave Sleep”

Delta (< 4 Hz)

3=<50% 4=>50%

REM sleep: low voltage-random, fast with sawtooth waves



REM

Variable EEG: 12-28 Hz

Desynchronized

Synchronized

Desynchronized

Awake

Beta (18-24 Hz)

Alert & Relaxed

Alpha (8 -12 Hz)

Stage 1

Theta (4-7 Hz)

Stage 2:

Theta w/Spindles
and K-Complexes

Stage 3 & 4

“Slow Wave Sleep”

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REM

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Desynchronized

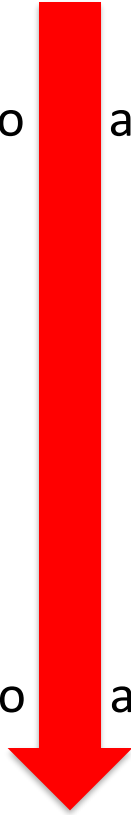
Lob many stones into a pond...

Lob one stone into a pond...

Synchronized

Lob *imaginary* stones into a pond...

Desynchronized



Awake



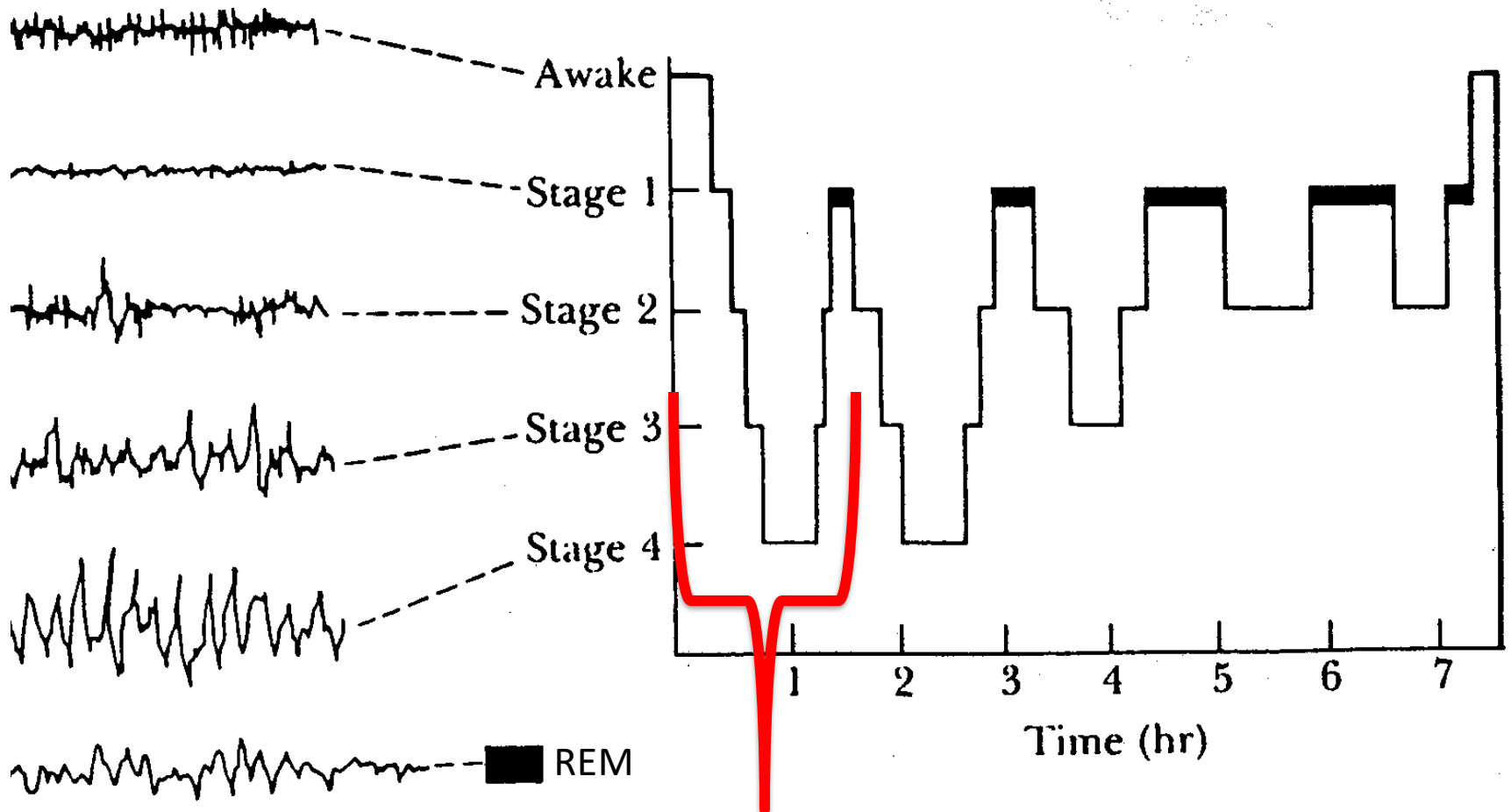
Stage 3 & 4
“Slow Wave Sleep”



REM

Sleep Cycle

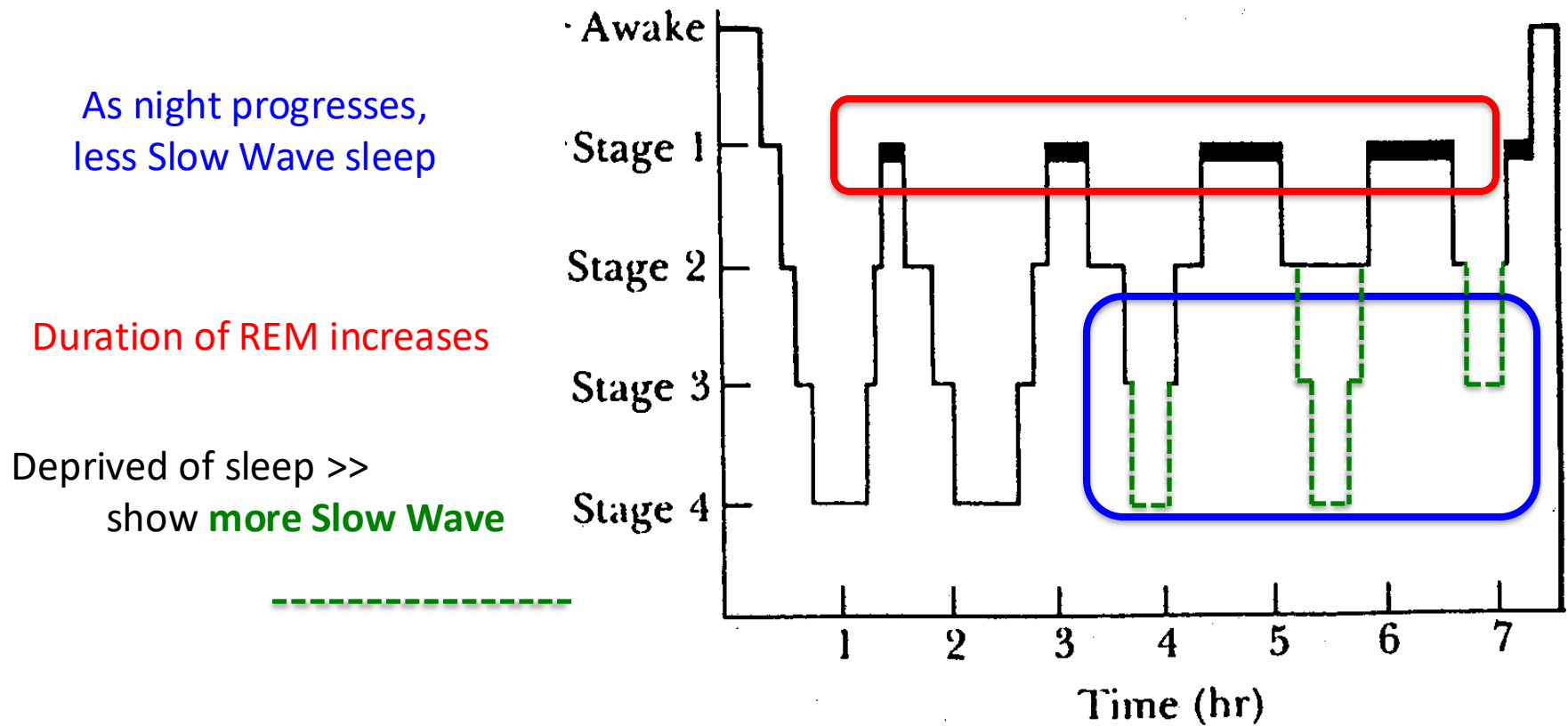
~ 90 Minute Cycle: Stage 1, 2, 3, 4, 3, 2, REM



~90 Min

Sleep Cycle

~ 90 Minute Cycle: Stage 1, 2, 3, 4, 3, 2, REM



Sleep Deprivation >> Lethargy, poor concentration, irritability;
Decreased resistance to infection

REM (Rapid Eye Movement) Sleep

Also called “Paradoxical” Sleep

- EEG Desynchronized
high freq, low voltage
- Heart & breathing rates
highly variable
- Eyes move, genitals active
but muscles paralyzed
“ATONIA”
- Highly correlated with
DREAMING



REM Deprivation

Cat on pedestal in tank – recording EEG



When enters REM,
Atonia causes cat to fall in,
wake up

So, selectively deprive it of REM sleep

REM-deprived cat displays....

Irritability

Poor concentration

Hallucinations

Death!



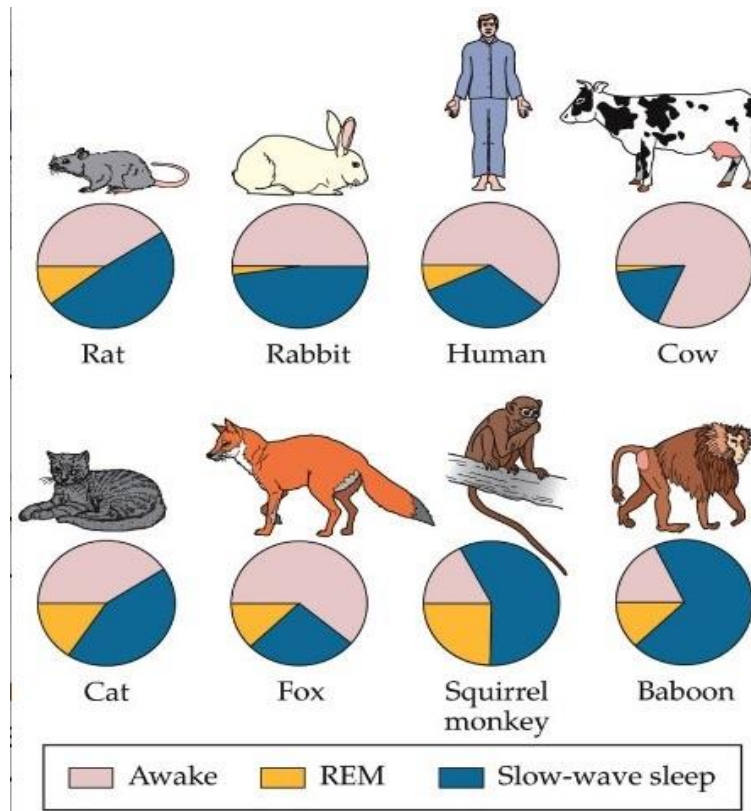
If REM deprived, will show

REM rebound

- REM periods lengthen

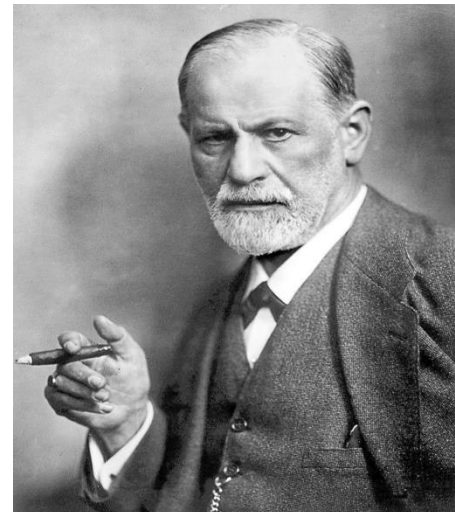
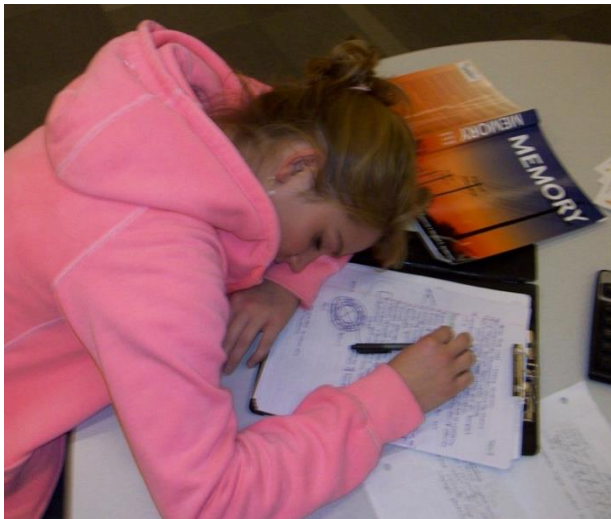
Functions of Sleep & REM ???

- Controversial!
- Sleep IS restorative
 - Engage in critical metabolic processes, process waste, etc
 - But, then why do some animals (e.g. prey) sleep only a few hours while others require so much more (e.g. predators) ?



Functions of Sleep & REM ???

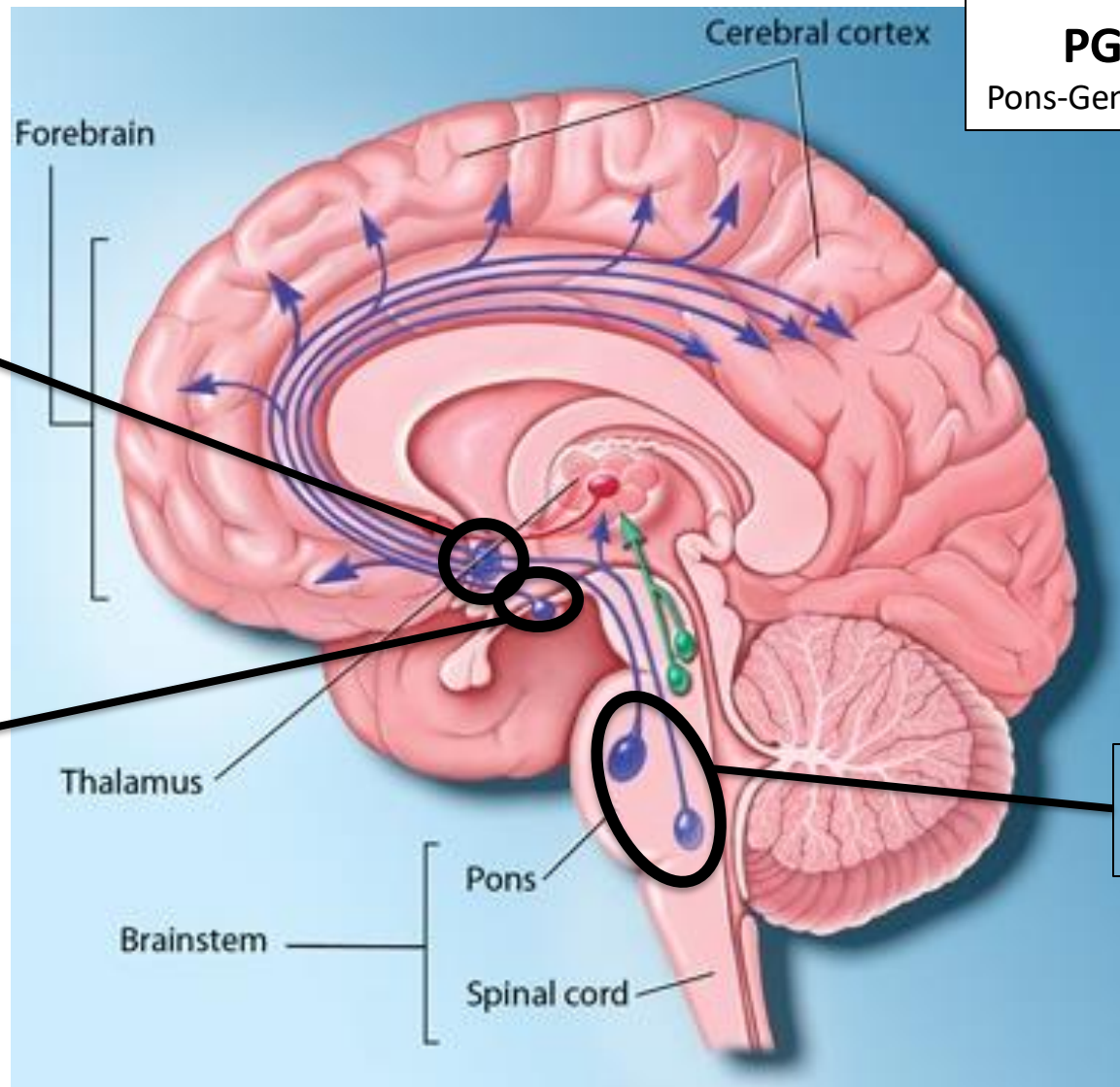
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 - REM-deprived have poorer memory for previous day's learning
 - Dreams may be involved in helping to resolve psychological conflicts



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- Slow-Wave/REM cycle may also involve temperature regulation
 - Brain operates within a very narrow range of temperatures
 - Brain cools down during Slow-Wave sleep
 - REM warms brain (Poss why Hypothalamus triggers PGO wave?)

Regulating the Sleep Cycle



Hypothalamus:
VLPOA
(Vento-Lateral
Pre-Optic Area)
>>**Basal Forebrain**

Hypothalamus:
**Tubero-
Mammillary**

Plus
PGO Wave
Pons-Geniculate-Occipital

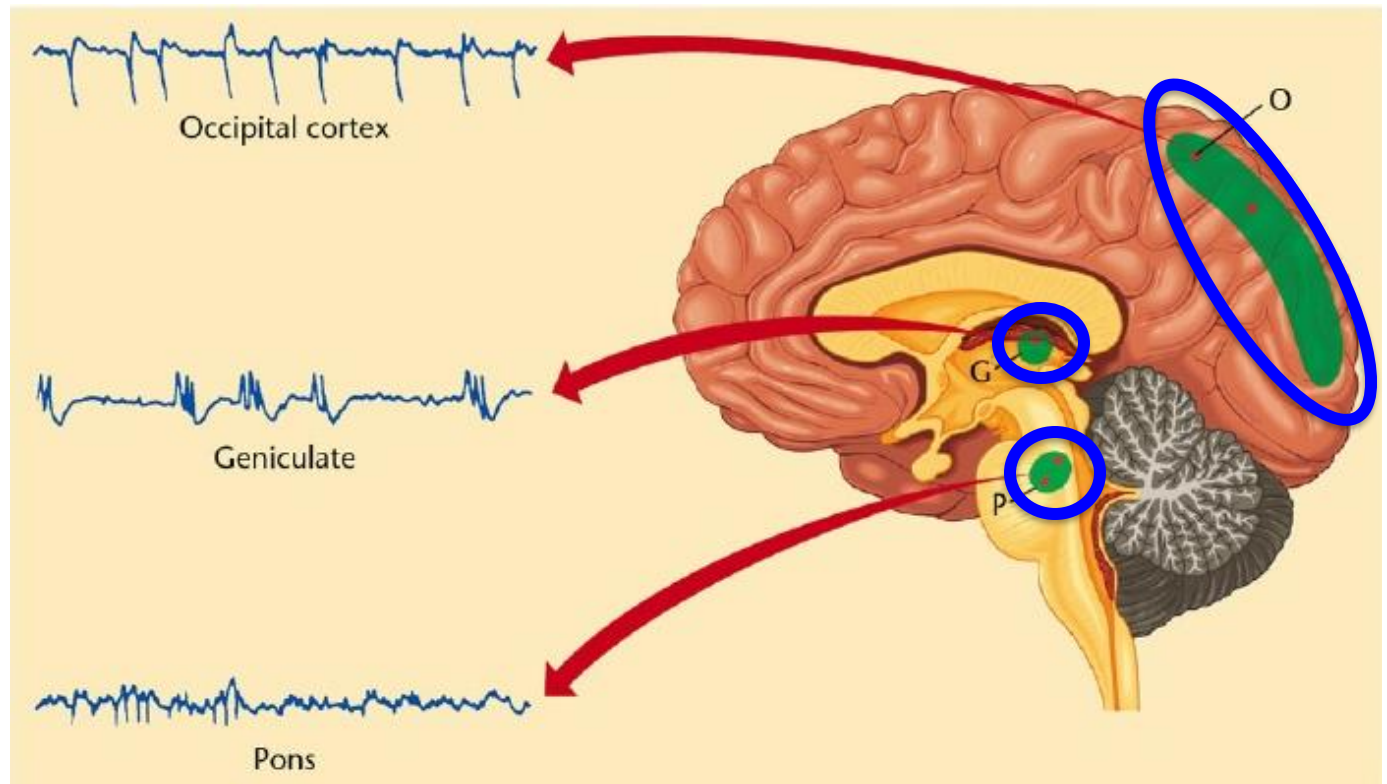
Raphe Nuclei
in medial Pons

REM – Initiated by PGO Waves

Then to
OCCIPITAL
(Visual) Cortex

Then to
Lateral Geniculate
of THALAMUS

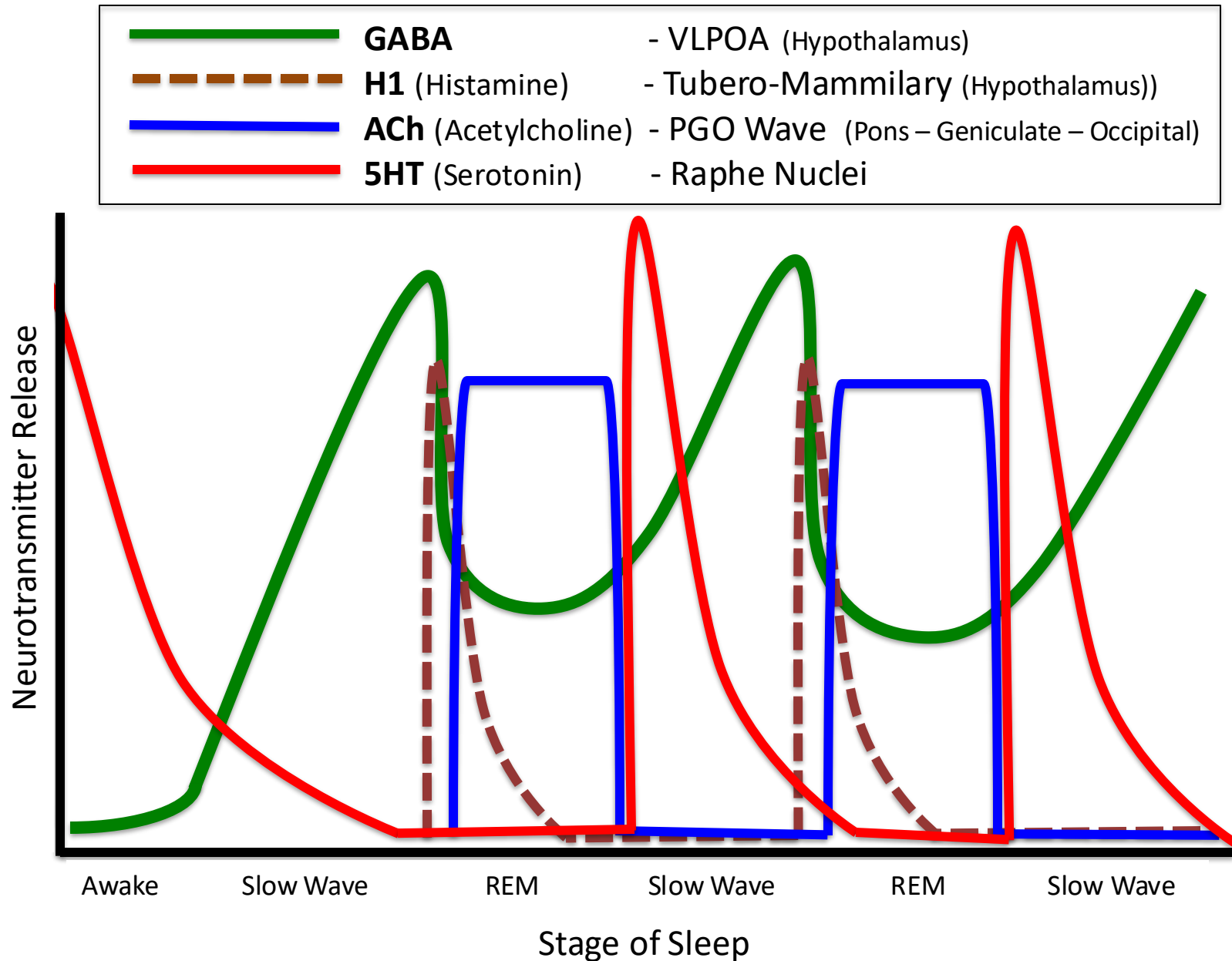
Activity begins in
PONS



REM

P → G → O

(Some) Neurotransmitters regulating Sleep Cycle



Circadium Rhythm

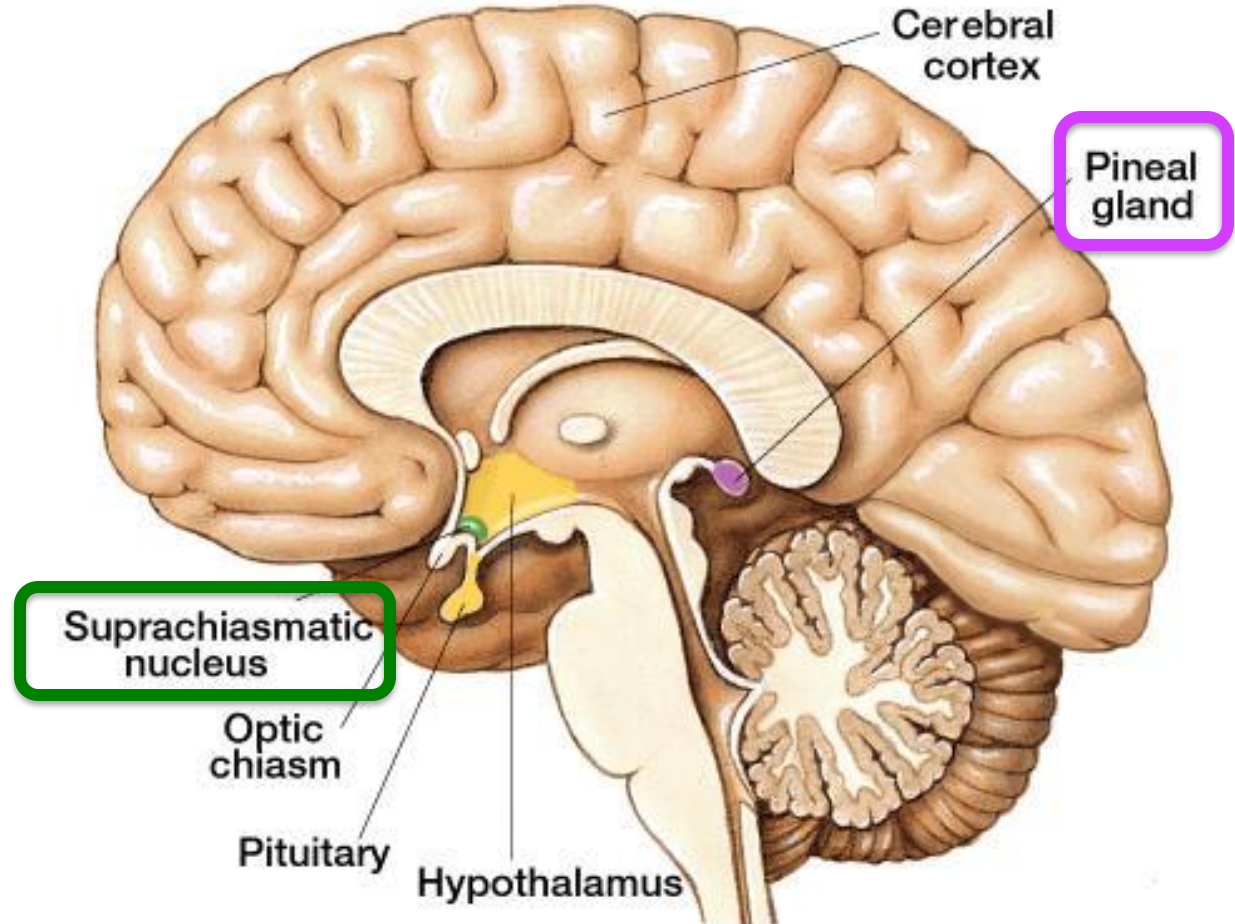
SCN

Suprachiasmatic Nucleus
of Hypothalamus

"Clock"
maintains a
24 +/- 1 hour
rhythm of activity

Virtually impossible to
disrupt this rhythm
in living cells

Apparently
genetically controlled



Interacts with **Pineal Gland** – see more below

Circadium Rhythm

— Subject asleep
— Subject awake

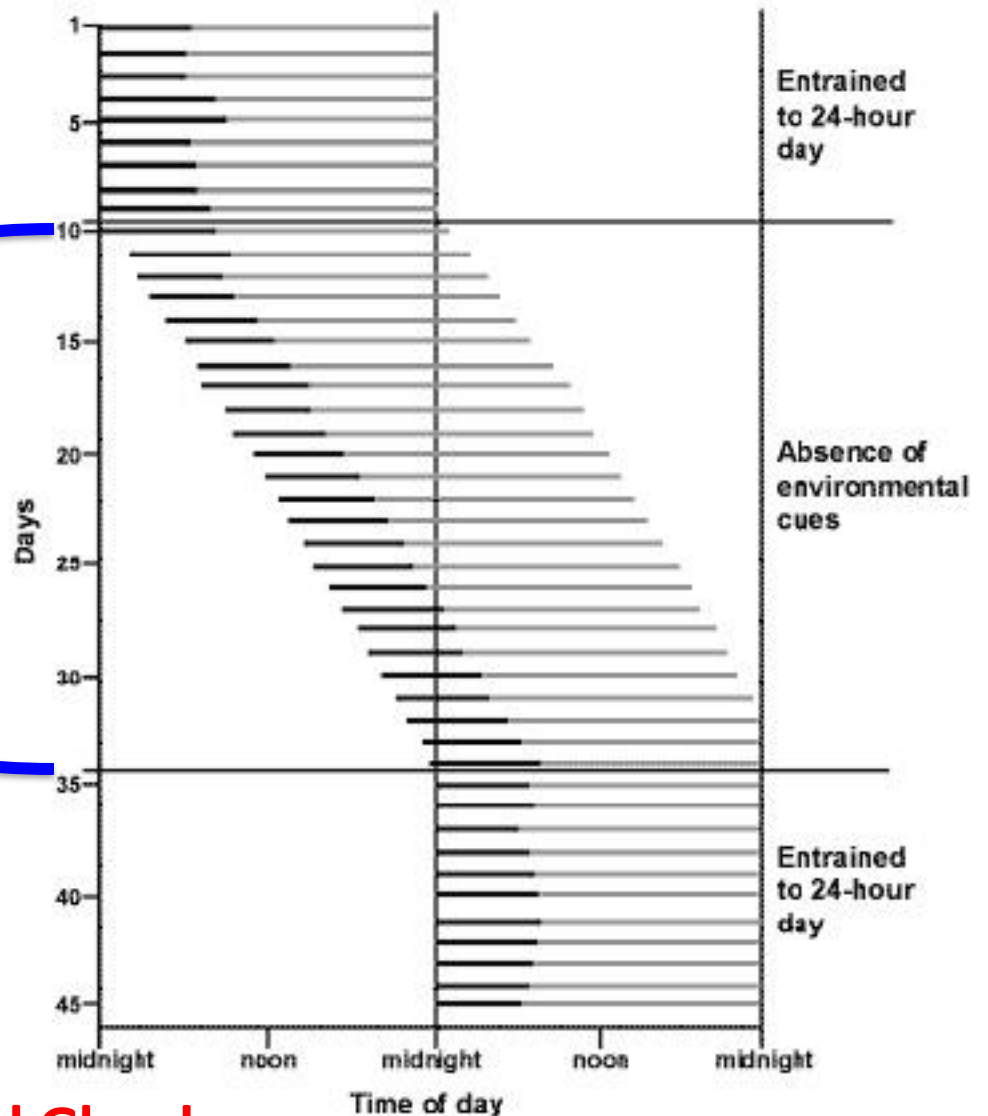
In the absence of
Day/Night cues...

...subject maintains
24= $\pm$ 1 hour rhythm...

...but cycle tends to **drift**

This flexibility allows creature
to adapt to seasonal changes.

So, in addition to **SCN** clock,
need a way to link it to
time of day



"Zeitgeber" (Time Giver) = **Pineal Gland**

The Pineal Gland

"The Third Eye"

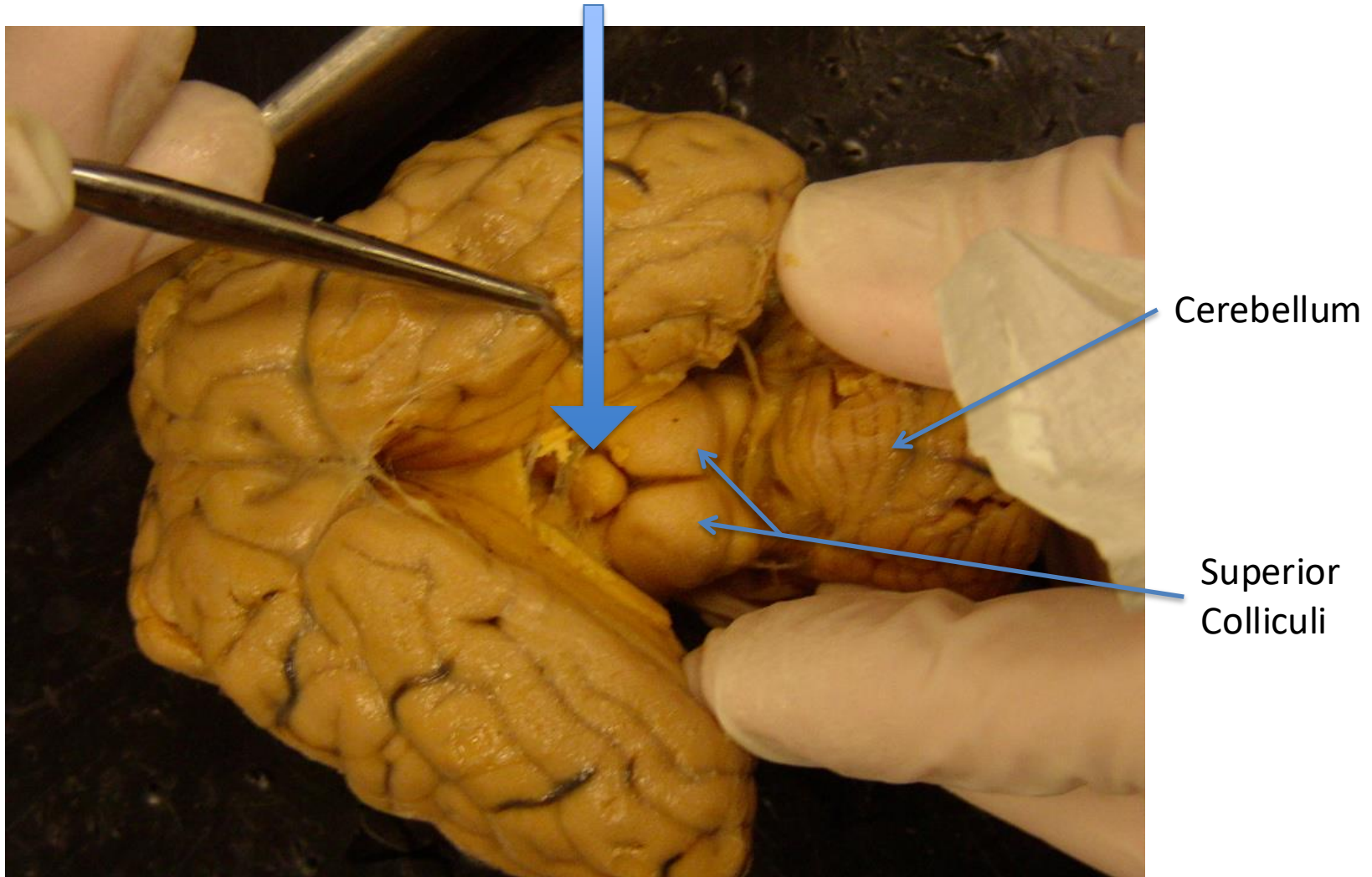


In some lizards & birds,
the **Pinal Gland**
(or specialized visual receptors to the Pineal Gland)
is a third eye!



While not exposed to the outside in humans,
it serves much the same function . . .

The Pineal Gland



Pineal gland releases **Melatonin** – promotes Sleep

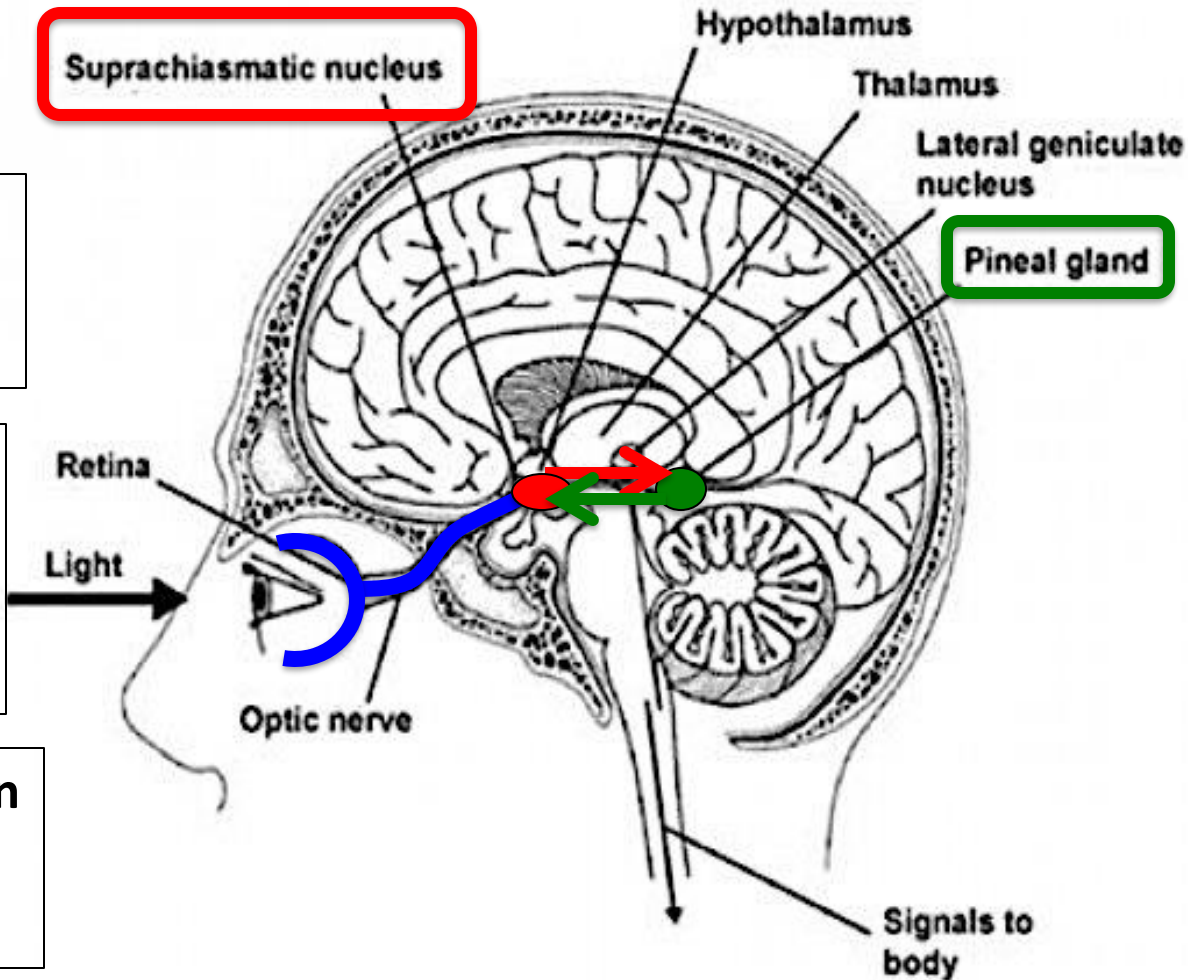
Retino-Hypothalamic Path

Specialized visual receptors
(ancient **Ganglion** cells,
with photopigment **Melanopsin**)
react to ambient light levels

These Ganglions send their axons
not to LGN, but to **SCN**
of Hypothalamus

SCN informs **Pineal Gland**,
so daylight decreases/
nightfall increases its release of
Melatonin

Pineal Gland releases **Melatonin**
inhibiting **SCN**
regulating its active/inactive cycle



NOTE – You can take Melatonin for jet-lag

Fly east, take about 1 hour before bed, promotes Pineal production of Melatonin, resets "clock"